

Polynomials with Predicates: a Dialectica reading*

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Abstract

A Dialectica object is a set of witnesses, a set of challenges and a relation saying which witness withstands which challenge, with morphisms running forward on the witnesses and backward on the challenges. Forgetting the relation leaves a polynomial functor of a special kind, one with a set of directions attached to each position. Dialectica Categories (1991) give two constructions of this shape, differing over whether the backward map may see the witness or not. Under this reading the Dialectica connectives, three tensors, two implications, and the additives, are operations that the theory of polynomial functors has since renamed, with the exception of *par* the multiplicative disjunction. Further work on Dialectica uncovered 'sequential' connectives, inspired by linear models of state, that also have polynomial shapes. We can easily put a predicate on the substitution product of polynomials, and this gives a new Dialectica sequencing connective in which the witness of the second component is chosen in the light of the challenge to the first. The two tensors are then related by a single map, the middle interchange, and everything else follows from it: a context may be given more order than it has, the sequential modality is determined rather than postulated, and whatever may be reused in parallel may be reused in sequence. This is how combining things in parallel and combining them in order are made to fit together. And, if you like the label, this is what a normal duoidal category buys. The Dialectica construction carries the structure.

*This note grew out of the 2022 Mathematics Research Community on applied category theory, where one of the working groups looked at the relationship between the Dialectica construction and polynomials. That group's own account of it is Dorta, Jarvis and Niu [8].

1 Dialectica constructions

The Dialectica construction reads a proposition as a two-move dialogue. The proposition is presented by a set U of witnesses, a set X of counterexamples or challenges, and a relation α saying which witness withstands which challenge, so that the proposition asserts $\exists u \in U. \forall x \in X. \alpha(u, x)$: some witness withstands every challenge to it. To pass from one proposition to another is to produce a witness for the second from a witness for the first, and to turn every challenge that might be raised against what was produced back into a challenge to what one started with. So the data of a morphism is a pair of maps running in opposite directions, one forward on witnesses and one backward on counterexamples, together with a condition saying that the backward map turns challenges back well.

Definition 1 (**Dial**($\mathcal{C}$), 1991). Let $\mathcal{C}$ be a cartesian closed category with finite products. An object of the category **Dial**($\mathcal{C}$) is a triple $A = (U, X, \alpha)$ with U and X objects of $\mathcal{C}$ and α a subobject of $U \times X$, read as saying which witness $u \in U$ withstands which counterexample $x \in X$. A morphism of **Dial**($\mathcal{C}$) from A to $B = (V, Y, \beta)$ is a pair

$$f: U \longrightarrow V, \quad F: U \times Y \longrightarrow X$$

in $\mathcal{C}$ such that $\langle \pi_1, F \rangle^* \alpha \leq (f \times 1_Y)^* \beta$ as subobjects of $U \times Y$. In $\mathcal{C} = \mathbf{Set}$ this says that $\alpha(u, F(u, y))$ implies $\beta(fu, y)$: if u withstands the challenge $F(u, y)$ that y was turned into, then fu withstands y . Composition is $(g, G) \circ (f, F) = (gf, (u, z) \mapsto F(u, G(fu, z)))$.¹

The comparison below is with a construction on sets, so $\mathcal{C}$ is **Set** from here on and α is an ordinary relation. The Dialectica construction comes in two forms, both introduced in [10]. They differ over what the backward map is allowed to see. In the form just defined, F takes the witness u as well as the challenge y , so the challenge it hands back may be chosen differently for different witnesses; this is the Gödel Dialectica[4]. In the other form F is a single function $Y \rightarrow X$, fixed once and for all, and this is the Girard Dialectica [9], written **Gir**(**Set**) here.

The shape of that data, forward on one component and backward on the other, is not peculiar to the Dialectica construction, and it is worth looking at where else it occurs before returning to the predicates.

¹The hypothesis is stronger than the category needs: finite products are enough to define **Dial**($\mathcal{C}$), and the cartesian closure is used for the structure carried by it in this note.

2 Polynomials and monomials

A polynomial functor is a set together with a set at each of its elements. The elements are called *positions* and the sets attached to them are the sets of *directions* available at each position, so the datum is a set I and a family $(X_i)_{i \in I}$: the same datum as a container in [3], or as an interface at which one may be in any of the positions i and may then choose among the directions in X_i . It is a functor because it acts on a set S by $S \mapsto \sum_{i \in I} S^{X_i}$, but nothing below uses that action; positions and directions are what we need. The category they form is called **Poly** [11].

A morphism of polynomials goes forward on positions and backward on directions. Given $(I, (X_i))$ and $(J, (Y_j))$, it consists of a function $\varphi_1: I \rightarrow J$ and, for each position i , a function $\varphi_i^\sharp: Y_{\varphi_1 i} \rightarrow X_i$: having sent the position i to the position $\varphi_1 i$, it must send each direction available there back to a direction available at i . The backward function is indexed by the position, so it may differ from one position to another.

The notation is the reason for the name. Write y^X for the polynomial with a single position carrying the directions X , and add such terms to put positions side by side, so that

$$p = \sum_{i \in I} y^{X_i}$$

is the polynomial above; a polynomial with three positions, two of them carrying three directions and one carrying one, is $2y^3 + y$. We write $p(1)$ for its set of positions and $p[i]$ for the directions at i .

Given this terminology *monomials* are where the comparison with the Dialectica construction lives, and the reason is the quantifier. A Dialectica object asserts $\exists u \in U. \forall x \in X. \alpha(u, x)$, and the $\forall$ ranges over the same X whatever witness the $\exists$ produces: the challenges available to the opponent are fixed in advance, and only the predicate tells one witness from another. A general polynomial lets the second set vary with the first, $\exists i \in I. \forall x \in X_i$, which is a proposition of a different shape. So the monomials are the polynomials that present propositions in the Dialectica sense, and an arbitrary polynomial is what one gets by letting the challenges depend on the witness.

Definition 2 (monomial). A polynomial is a *monomial* when the directions do not vary with the position: $p = Uy^X$, with $p(1) = U$ and $p[u] = X$ for every u . We write **Poly**_{mono} for the full subcategory of monomials.

Writing the same set X at every position is a matter of how the polynomial is presented. What matters is the weaker condition that the sets at

the positions are all isomorphic, since a bijection at each position gives an isomorphism $y^X \cong y^{X'}$ whenever $X \cong X'$. So a polynomial whose direction sets agree up to isomorphism is already a monomial, and fixing one set X is a choice of notation, exactly as fixing the set of counterexamples of a Dialectica object is, and $\mathbf{Poly}_{\text{mono}}$ is closed under isomorphism in the sense that matters.

Monomials are the basic building blocks of polynomials: the representables y^X are the blocks, every polynomial is a sum of them, and a monomial is a sum of copies of one. That a class picked out on such elementary grounds should be the one carrying logical interpretations is suggestive.

3 Dialectica and monomials

A morphism between monomials $Uy^X \rightarrow Vy^Y$ is a function $f: U \rightarrow V$ together with a U -indexed family of functions $Y \rightarrow X$, which is a single function $F: U \times Y \rightarrow X$. So a morphism of monomials is exactly the underlying data of a Dialectica morphism. Call the monomial Uy^X the *shape* of the Dialectica object (U, X, α) : what is left of it when the predicate is forgotten. Write $|-|: \mathbf{Dial}(\mathbf{Set}) \rightarrow \mathbf{Poly}_{\text{mono}}$ for the functor taking each object to its shape and each morphism to its underlying pair, which is faithful.

The two theorems below are what the rest of this note rests on. Their proofs are a matter of reading definitions, but what they say is that every connective of the two constructions splits into a polynomial operation on the shapes and an operation on the predicates, and that the shapes stay inside the monomials throughout. Once the two parts are separated they can be varied independently: the shape can be changed while the predicate is kept, or the shape kept while the predicate takes its values somewhere other than in the boolean. So the constructions stop being a list of definitions to be checked one at a time and become a landscape of variations to be surveyed, which is what the sections below do.

Theorem 3 (connectives of the Gödel Dialectica). *Let $A = (U, X, \alpha)$ and $B = (V, Y, \beta)$ be objects of $\mathbf{Dial}(\mathbf{Set})$. The connectives of $\mathbf{Dial}(\mathcal{C})$ (from [10]) read as follows in the canonical case $\mathcal{C} = \mathbf{Set}$, and the shape of each is a monomial.*

1. *The Dialectica tensor $A \otimes B$ has witnesses $U \times V$ and challenges $X \times Y$: a witness is a witness of each component and a challenge is a challenge to each, so $(U \times V, X \times Y, \wedge)$. Its unit is $I = (1, 1, \text{id})$, of shape y . The shape $(U \times V)y^{X \times Y}$ is what $\mathbf{Poly}$ calls the Dirichlet product.*

2. The Dialectica linear implication $A \multimap B$ has for witnesses the pairs of maps $(f: U \rightarrow V, F: U \times Y \rightarrow X)$, which is the data of a morphism from A to B with the condition on predicates not yet imposed, and for challenges the pairs (u, y) of a witness of A and a challenge to B , so $(V^U \times X^{U \times Y}, U \times Y, \beta^\alpha)$. A witness (f, F) withstands (u, y) when u withstanding $F(u, y)$ implies fu withstanding y , so the witnesses that withstand every challenge are exactly the morphisms $A \rightarrow B$. It is the closure of the original dialectica tensor:

$$\mathbf{Dial}(\mathbf{Set})(C \otimes A, B) \cong \mathbf{Dial}(\mathbf{Set})(C, A \multimap B).$$

Its shape $(V^U \times X^{U \times Y})y^{U \times Y}$ is the closure that **Poly** has for the same product, where the corresponding adjunction holds for every polynomial C , monomial or not.

3. The Dialectica additive conjunction $A \& B$ has witnesses $U \times V$ and challenges $X + Y$, so $(U \times V, X + Y, \text{choose})$: a challenge picks one component and challenges it. Its unit is $(1, 0)$, of shape y^0 . The shape $(U \times V)y^{X+Y}$ is the cartesian product in **Dial(Set)** and in **Poly**.
4. The additive disjunction $A \oplus B$ has witnesses $U + V$ and challenges $X \times Y$, a witness of one component being challenged by the corresponding component of the pair. It receives both injections, but it is a weak coproduct only in **Dial(Set)**, while it is a coproduct in **Gir(Set)**: there the copairing of (f, F) and (g, G) has backward part $\langle F, G \rangle$, which is available because a backward map is a single function, whereas here a backward map out of $A \oplus B$ is defined on $(U + V) \times Z$ and has nothing to supply its Y component at a witness from U .

Proof. Items (1) and (3) are the two formulas read off at constant directions: the Dirichlet product multiplies positions and multiplies directions, and the cartesian product multiplies positions and adds directions. Item (4) is read off in the same way. For (2), let $C = \sum_{w \in W} y^{Z_w}$ be arbitrary. Then $C \otimes A = \sum_{(w, u)} y^{Z_w \times X}$, so a morphism $C \otimes A \rightarrow B$ consists of $g: W \times U \rightarrow V$ together with h_1 sending (w, u, y) into Z_w and $h_2: W \times U \times Y \rightarrow X$. Currying in W turns g into $W \rightarrow V^U$ and h_2 into $W \rightarrow X^{U \times Y}$, and leaves h_1 as a family of functions $U \times Y \rightarrow Z_w$ indexed by w . That is exactly a morphism $C \rightarrow (V^U \times X^{U \times Y})y^{U \times Y}$, naturally in C . $\square$

The second construction, where the backward map is a single function $Y \rightarrow X$, has connectives of its own, and the same holds of them.

Theorem 4 (connectives of the Girard Dialectica). *Let A and B be dialectica objects as above, now taken in $\mathbf{Gir}(\mathbf{Set})$.*

1. *The cross product $A \otimes_{\mathbf{Gir}} B$ has witnesses $U \times V$ and for challenges the pairs of functions $(f: V \rightarrow X, g: U \rightarrow Y)$, each choosing a challenge to one component in the light of the witness of the other; a pair (u, v) withstands (f, g) when u withstands fv and v withstands gu , so $(U \times V, X^V \times Y^U, \text{cross})$. Its unit is I again. Its shape is $(U \times V) y^{X^V \times Y^U}$.*
2. *The cross product is closed, the implication having witnesses the pairs $(f: U \rightarrow V, F: Y \rightarrow X)$ and challenges $U \times Y$, with (f, F) withstanding (u, y) when u withstanding Fy implies fu withstanding y , so $(V^U \times X^Y, U \times Y, \beta^\alpha)$. Its shape is $(V^U \times X^Y) y^{U \times Y}$; the adjunction is taken with respect to the morphisms of this construction, whose backward maps do not see the witness.*
3. *The coproduct $A \oplus B$ has witnesses $U + V$ and challenges $X \times Y$, so $(U + V, X \times Y, \text{choose})$, of shape $(U + V) y^{X \times Y}$. Here it is a genuine coproduct, for the reason given in Theorem 3(4).*
4. *The par $A \wp B$ has for witnesses the pairs of functions $(f: Y \rightarrow U, g: X \rightarrow V)$ and for challenges the pairs (x, y) , with (f, g) withstanding (x, y) when fy withstands x or gx withstands y , so $(U^Y \times V^X, X \times Y, \vee)$. Its shape is $(U^Y \times V^X) y^{X \times Y}$, which puts function spaces among the positions and a plain product among the directions; we have not found that combination among the monoidal structures listed in [12].*

Proof. Each item is the corresponding definition of [10] read at the level of shapes, the backward maps of this construction being single functions rather than families. $\square$

The polynomial connectives on monomials are the Dialectica ones, and they were written down thirty years earlier. The predicate is not something the items above add to them: it was there from the start, and it is what the functor $|-|$ forgets. Each connective computes it alongside the shape, $\alpha \wedge \beta$ pointwise for the tensor, $\alpha(u, F(u, y)) \Rightarrow \beta(fu, y)$ for the implication, and α on the left summand with β on the right for the conjunction, so that $|-|$ carries each connective to the corresponding polynomial structure and the adjunction $\otimes \dashv \multimap$ in $\mathbf{Dial}(\mathbf{Set})$ is the adjunction of item (2) together with the residuation $\gamma \wedge \alpha \leq \beta$ iff $\gamma \leq (\alpha \Rightarrow \beta)$ applied pointwise.

The shape of the cross product (the tensor product of $\mathbf{Gir}(\mathbf{Set})$) appears in $\mathbf{Poly}$ without a name, among the monoidal products collected in [12].

3.1 More tensors

The Dialectica objects allow many more monoidal bifunctors. Blass worked with the objects of $\mathbf{Gir}(\mathbf{Set})$, which he calls $\mathcal{PV}$, and defined on them a further tensor product [1]. He arrives at it from the set-theoretic reading of the morphisms, where a proof of an inequality between three cardinal characteristics is a morphism out of a composite of two objects, and the composite asks a first question, receives an answer, and chooses the second question in the light of that answer.

Definition 5 (Blass). Given A and B as before, written with the two sides exchanged so that it is the challenge to the first component that is chosen,

$$A \rtimes B = (U \times V, X^V \times Y, \alpha(u, fv) \wedge \beta(v, y)),$$

a witness being a pair (u, v) and a challenge a pair (f, y) with $f: V \rightarrow X$, which withstand each other when u withstands fv and v withstands y .

The witnesses are a plain product, but the challenges have one exponential, X^V in place of X . The same operation is one of the two further monoidal products listed at the end of [8]. On homogeneous polynomials over the booleans, the one with function spaces on both sides is the cross product of Theorem 4(1), the multiplicative conjunction of the Girard Dialectica, where it interprets $\otimes$ and stands with the par in a de Morgan pair [9, 10]; the one with a function space on a single side is Blass's, interpreting a reduction that asks a question, reads the answer, and asks a second in the light of it [1].²

The definition of Blass' tensor has a mirror image, and this is the connective that **Poly** has and Dialectica never considered before, although it knew the connective was there. What told it so is a map it already had. The two tensors of the thesis are related by

$$A \otimes B \longrightarrow A \otimes_{\mathbf{Gir}} B,$$

²Dorta, Jarvis and Niu [8] give both products in their closing section and ask whether interpretations or applications can be found for them. Both have had one for more than thirty years, and in the subject their paper is about. The two-sided product is the tensor of the category GC , the multiplicative conjunction of a model of classical linear logic, given in [9] and in the last two chapters of [10], and the category was built so that this connective should mean that. The one-sided product is the sequential composition of [1], where it is what a proof of an inequality between three cardinal characteristics is a morphism out of. Nor is the identification remote: [8] identifies the homogeneous polynomials over the booleans with **Dial**(**Set**), citing [10] for the construction.

the identity on witnesses, sending a challenge (f, g) of the cross product back to the challenge (fv, gu) at the witness (u, v) : the constructions already contained a passage from no dependence at all to dependence on both sides, in the challenge coordinate, and Blass’s tensor is the step halfway along it. Linear negation carries the cross product to the par, whose witnesses are $U^Y \times V^X$, so the same passage exists in the witness coordinate, from $\otimes$ to $\wp$. Its halfway step was never written down, and what was missing was not the shape but a reading for it: a shape becomes a connective when something says what it means, and what says so for this one is the calculus of Section 6, whose ordered contexts fix what “before” has to satisfy. Instead of letting a challenge be chosen in the light of the other component’s witness, let a witness be chosen in the light of the other component’s challenge.

Proposition 6. *Given A and B dialectica objects, consider the bifunctor*

$$A \triangleleft B = (U \times V^X, X \times Y, \alpha \triangleleft \beta), \quad (1)$$

whose witnesses are a witness of A together with a choice of witness of B for each challenge to A , and whose challenges are a challenge x to the first component paired with a challenge y to the witness that x elicited. It is defined on the objects of $\mathbf{Gir}(\mathbf{Set})$, as $\times$ is, and equally on those of $\mathbf{Dial}(\mathbf{Set})$, the two constructions having the same objects. On shapes it is the substitution product of $\mathbf{Poly}$.

Proof. A witness of the composite is a witness u of A together with a rule φ choosing, for each challenge x that may be raised against u , a witness of B to be confronted next; so the witnesses are $U \times V^X$. A challenge to it names the challenge x to be raised first and a challenge y to whatever x elicited, and since the challenges to B do not vary with its witness that second choice is made from Y , so the challenges are $X \times Y$. Given $(f, F): A \rightarrow A'$ and $(g, G): B \rightarrow B'$, the witness (u, φ) is carried to $(fu, x' \mapsto g\varphi F(u, x'))$ and the challenge (x', y') back to $(F(u, x'), G(\varphi F(u, x'), y'))$, which is functorial in each argument; the condition on the predicates is checked in Section 4, where the relation $\alpha \triangleleft \beta$ is given. Forgetting the predicates, the witnesses and challenges just computed are the positions and directions of the substitution product of the two monomials. $\square$

The relation $\alpha \triangleleft \beta$ is written out in Section 4, where the values are allowed to lie in a lineale, since the two predicates are combined there in a fixed order. Table 1 collects the connectives of the two constructions together with the two of this section, and names the polynomial operation carried by each shape.

	witnesses	challenges	Poly
<i>in Dial(Set)</i>			
tensor $A \otimes B$	$U \times V$	$X \times Y$	Dirichlet product
implication $A \multimap B$	$V^U \times X^{U \times Y}$	$U \times Y$	its closure
conjunction $A \& B$	$U \times V$	$X + Y$	cartesian product
disjunction $A \oplus B$	$U + V$	$X \times Y$	—
<i>in Gir(Set)</i>			
tensor $A \otimes_{\mathbf{Gir}} B$	$U \times V$	$X^V \times Y^U$	the chain (178)
implication $A \multimap_{\mathbf{Gir}} B$	$V^U \times X^Y$	$U \times Y$	—
conjunction $A \& B$	$U \times V$	$X + Y$	cartesian product
disjunction $A \oplus B$	$U + V$	$X \times Y$	—
par $A \wp B$	$U^Y \times V^X$	$X \times Y$	—
<i>further connectives</i>			
Blass's $A \bowtie B$	$U \times V$	$X^V \times Y$	the chain (178)
substitution $A \triangleleft B$	$U \times V^X$	$X \times Y$	substitution product

Table 1: The connectives of the two Dialectica constructions, with Blass’s tensor and the substitution tensor, given by their witnesses and challenges and by the polynomial operation on their shapes. A dash means we have not found the operation in the monoidal polynomial list in [12].

Not every operation of **Poly** appears in the table. Some leave the monomials, and they leave them for one reason: they let the directions vary with the position. The free monad on Uy^X has trees for positions and the leaves of each tree for directions, which differ from one tree to the next; the derivative removes a direction, so what remains depends on which was removed; and the coproduct of **Poly** is the disjoint union $Uy^X + Vy^Y$, which keeps each summand’s own directions. The dialectica constructions have two additive disjunctions and neither is that one. One is the weak coproduct in **Dial(Set)**, where a backward map out of it has nothing to supply at a witness from the wrong summand, and a genuine coproduct in **Gir(Set)**, where the backward map is a single function and the copairing is available. So the disjoint union of polynomials is a third thing, and it is the one that leaves the monomials. Others stay inside the monomials and are Dialectica objects already. What a predicate adds to them is not taken up here.

The Dialectica reading is not a proposal to work with polynomials with predicates. The predicate is one dial and the shape is another, and the shape has settings this note leaves alone. The challenges may be fixed, which is the Dialectica setting and the one used throughout; they may vary with the witness, which is the general polynomial; the witnesses may vary with the challenge, which is the same dependence read in the other direction; or

neither may be fixed, which asks for two indices rather than one. Which operations work in each setting is what a comparison is for, and only the first setting is treated below.

4 Substitution tensor

From here on fix a lineale L and write $\mathbf{Dial}_L(\mathbf{Set})$ for the category of Definition 1 with $\alpha: U \times X \rightarrow L$ and the morphism condition $\alpha(u, F(u, y)) \sqsubseteq \beta(fu, y)$; Theorem 3 is about shapes and is unaffected, and $\mathbf{Dial}(\mathbf{Set})$ is the case $L = 2$. The lineale is what the predicate of the substitution tensor needs, since two predicates are combined there in a fixed order.

In this section we want to explain the substitution tensor in Dialectica terms, so Equation (1) gives the shape and here is the predicate. A witness of $A \triangleleft B$ is a witness u of A together with a function φ assigning a witness of B to each challenge to A ; a challenge to it is a pair (x, y) , a challenge x to u together with a challenge y to the witness φx that x elicited. So $A \triangleleft B$ is A and then B , with the second component's witness chosen in the light of how the first was attacked, and with the two challenges made in that order.

Definition 7 ($\triangleleft$ tensor). For dialectica objects $A = (U, X, \alpha)$ and $B = (V, Y, \beta)$ set

$$\begin{aligned} A \triangleleft B &:= (U \times V^X, X \times Y, \alpha \triangleleft \beta), \\ (\alpha \triangleleft \beta)((u, \varphi), (x, y)) &:= \alpha(u, x) \otimes \beta(\varphi x, y), \end{aligned}$$

the tensor being that of the lineale L . This is the Dialectica version of the *substitution tensor* of polynomials.

The device of putting a function space into one of the two coordinates of a Dialectica object is Blass's, and Definition 5 puts it on the challenges. What is taken up here is the other coordinate, together with the algebra that comes with it: a function space among the witnesses makes the operation on shapes the substitution product of polynomials, so that associativity, the unit and the interchange with $\otimes$ are properties of that operation rather than calculations.

This is the composition product of [8] at $\mathcal{C} = 2$, where the predicate is allowed to take its values in any monoidal category and the predicate of a composite is the monoidal product of the two; the shape is theirs and so is the predicate on it, and what Section 6 adds is the reading. No exponential is asked of that category: the positions and the directions are sets throughout, and V^X is the X -indexed product of copies of V taken among them. Over

a base other than **Set** it is a real demand, and it is one of the uses that the cartesian closure of Definition 1 is put to.

Proposition 8. *The connective $\triangleleft$ is a bifunctor on $\mathbf{Dial}_L(\mathbf{Set})$, associative and unital with unit $I = (1, 1, \top)$, and $|A \triangleleft B| = |A| \triangleleft |B|$. It is not commutative.*

Proof. On shapes this is substitution in **Poly**, so associativity and unitality hold there and it remains to check the predicates. Given $(f, F): A \rightarrow A'$ and $(g, G): B \rightarrow B'$, the morphism $(f, F) \triangleleft (g, G)$ acts by

$$\begin{aligned} (u, \varphi) &\longmapsto (fu, x' \mapsto g\varphi F(u, x')), \\ (x', y') &\longmapsto (F(u, x'), G(\varphi F(u, x'), y')), \end{aligned}$$

the first components being substituted one into the other as in **Poly**. Writing $v := \varphi F(u, x')$, the required inequality is

$$\alpha(u, F(u, x')) \otimes \beta(v, G(v, y')) \sqsubseteq \alpha'(fu, x') \otimes \beta'(gv, y'),$$

which follows from the two morphism conditions and monotonicity of $\otimes$ in L . Unitality of the predicate is $\alpha(u, x) \otimes \top = \alpha(u, x)$ and associativity is associativity of $\otimes$ in L . Non-commutativity is visible in the witnesses: $A \triangleleft B$ has $U \times V^X$ and $B \triangleleft A$ has $V \times U^Y$. $\square$

Remark 9. There are two ways for one component of a tensor to see the other, and they are not the same. In $A \triangleleft B$ the witness of the second component depends on the challenge to the first, so what is passed travels forward along the play. In the one-sided product $A \rtimes B$ of Definition 5 the challenge to the first component depends on the witness of the second, so it travels the other way. Sequencing needs the first of these; the cross product has the second on both sides at once.

4.1 Relating Tensors

With every connective now written as a Dialectica object, the tensors fall into two chains that leave from the same place. Start from the Dialectica tensor $\otimes$, where neither component sees the other. Let a challenge to one component be chosen in the light of the witness of the other, on one side only, and the challenges become $X^V \times Y$, which is Blass's $\rtimes$ of Definition 5; let it happen on both sides and they become $X^V \times Y^U$, which is the cross product of Theorem 4(1). Along this chain the witnesses stay a plain product. Now move the exponentials to the other coordinate instead. Let the witness of

one component be chosen in the light of the challenge to the other, on one side only, and the witnesses become $U \times V^X$, which is the substitution tensor of Equation (1); on both sides they become $U^Y \times V^X$, which is the par of Theorem 4(4). Along this chain the challenges stay $X \times Y$. The shapes are those of Table 1, and the steps are morphisms.

Proposition 10. *In $\mathbf{Dial}(\mathbf{Set})$ there are morphisms, natural in all arguments,*

$$A \otimes B \longrightarrow A \rtimes B \longrightarrow A \otimes_{\mathbf{Gir}} B,$$

$$A \otimes B \longrightarrow A \triangleleft B \longrightarrow A \wp B,$$

the first chain the identity on witnesses and the second the identity on challenges.

Proof. For the first chain, send the challenge (f, y) of $A \rtimes B$ back to the challenge (fv, y) of $A \otimes B$ at the witness (u, v) , and the challenge (f, g) of $A \otimes_{\mathbf{Gir}} B$ back to the challenge (f, gu) of $A \rtimes B$. In each case the predicate of the source is the predicate of the target at those arguments, so the condition holds with equality. For the second chain, send the witness (u, v) to (u, const_v) and the witness (u, φ) to $(\text{const}_u, \varphi)$, the challenges going back unchanged; the predicates again agree, the last step using that a conjunction implies the corresponding disjunction. Naturality in each argument is naturality of the corresponding maps of shapes. $\square$

The two chains are not alike in what they ask of the construction. Both maps of the first evaluate a function at the other component's witness, so they need the backward map of a morphism to see the witness, and they have no counterpart in $\mathbf{Gir}(\mathbf{Set})$. The first step of the second chain is the map called **ser** in Section 6.

Linear negation exchanges the two coordinates of an object, so it exchanges the two chains term by term: the cross product is dual to the par, Blass's $\rtimes$ is dual to the substitution tensor $\triangleleft$, and the Dialectica tensor, at the head of both, is carried to itself. The additives are monoidal structures too, but do not take part in either chain.

5 Modalities and Comonoids

The connectives so far are the multiplicatives and the additives, and the constructions carry two more. A modality says which formulas may be used again, and in a categorical model it is a comonad whose values carry comonoid structure for a tensor, the comultiplication being the map that

splits a formula into two copies. There is one modality for each of two of the tensors, $\otimes$ and $\triangleleft$, and the difference between them is a monoid. Reuse in parallel is unordered, and the comonoid interpreting it is required to be commutative, so the challenges of $!A$ form the free commutative monoid on X , which is the finite multisets on it. Reuse one after another is ordered, and no commutativity is asked of the comonoid, so the challenges of $\dagger A$ form the free monoid, which is the words on X . The earlier models make the same match from the other side: with a commutative connective for “before” the modality took the free commutative monoid, and the version with a non-commutative one took plain sequences [6, 7].

Definition 11 (modalities). For $A = (U, X, \alpha)$ put

$$!A := (U, M_{\text{fin}}X, !\alpha), \quad \dagger A := (U, X^*, \dagger\alpha),$$

where $M_{\text{fin}}X$ is the free commutative monoid on X , that is the finite multisets on it, and X^* is the free monoid, and where $!\alpha(u, m)$ and $\dagger\alpha(u, w)$ are the $\otimes$ -product in L of the values $\alpha(u, x)$ over the entries of m and of w respectively. Both act on maps by the functor applied to the backward component.

Neither modality is new, and neither is the structure they carry. Definition 11 is the $!$ of the thesis [10] together with the $\dagger$ that came with the sequencing connective of the models of state [6, 7]. What was established there is more than that they are comonads: each is a *linear exponential comonad*, a monoidal comonad whose values carry comonoid structure compatibly with the comonad and with the tensor, which is exactly what a model of a linear modality requires. The counit is given by the singleton and the comultiplication by the multiplication of the monad on the challenge side, in each case the identity on witnesses.

What is wanted here is not the comonad structure but the comonoid structure each of them carries, since that is what a modality must have if it is to interpret reuse: a formula that may be used twice has to come with a map splitting it in two, and that map has to be a comonoid comultiplication for the tensor by which the two copies are combined. Each of the two modalities has one, for a different tensor.

Proposition 12. *Let A be an object of $\mathbf{Dial}_L(\mathbf{Set})$. The following are comonoid structures in that category, for $\otimes$ and for $\triangleleft$ respectively.*

1. $!A$ is a comonoid for $\otimes$, with comultiplication the diagonal on witnesses and the sum of multisets backwards, and counit the empty multiset;

2. $\dagger A$ is a comonoid for $\triangleleft$, with comultiplication sending u to (u, const_u) and concatenating words backwards, and counit the empty word;
3. the abelianisation $X^* \rightarrow M_{\text{fin}} X$ is the backward part of a comonad morphism $\kappa: ! \rightarrow \dagger$, the identity on witnesses.

Proof. In each case the predicate condition is an equality rather than an inequality, because the predicate of $!A$ at a multiset and of $\dagger A$ at a word is the $\otimes$ -product over the entries, and concatenation, multiset sum and abelianisation all preserve that product. The underlying maps of shapes are comonoid structures in **Poly** for the corresponding products, so with the predicate conditions checked the structures are comonoids in **Dial_L(Set)** itself. $\square$

The proposition is not new either. Item (1) is in the thesis [10], where $!$ is shown to be a linear exponential comonad on the Dialectica category, and items (2) and (3) are Proposition 2 of [7], with the sequencing connective $\odot$ of that paper in place of $\triangleleft$. What is new here is only the change of tensor: the witnesses of $A \triangleleft B$ are not a plain product, so the comultiplication of $\dagger A$ has a forward part to supply as well as a backward one, and it is the constant map $u \mapsto \text{const}_u$ that supplies it.

The comonoids for $\triangleleft$ deserve a paragraph of their own, because they have a reading that the comonoids for $\otimes$ do not. A comonoid for the substitution product in **Poly** is precisely a small category, by a result of Ahman and Uustalu [5]: the positions are the objects, the directions at a position are the morphisms out of that object, the counit picks the identity at each object, and the comultiplication says where each morphism lands and how two of them compose. The identification is not special to **Poly**. A comonoid for $\triangleleft$ in **Dial_L(Set)** is a category enriched in the lineale: the shape conditions give the category, and the predicate conditions say that the identity at each object has weight $\top$ and that the weight of a composite is the $\otimes$ of the weights of its factors, which is the enrichment. This is the theorem of [8, Thm. 4.3], and at $L = 2$ it returns the plain statement. Here the categories have the same set of morphisms out of every object, since the challenges do not vary with the witness.

The correspondence is one of objects and not of morphisms. A map of $\triangleleft$ -comonoids goes forward on objects and, at each object, backward on the morphisms out of its image; that is not a functor but a cofunctor, also called a retrofunctor. The backward direction is the one a Dialectica morphism has as well, which is why the two notions sit together as comfortably as they do, and on the enriched side the maps are enriched cofunctors.

For $\dagger A$ the category is a simple one. Its objects are the witnesses of A , the morphisms out of u are the words of challenges to u , composition is concatenation, and the comultiplication sends every word back to the same object, so there are no morphisms between distinct witnesses. Read as a transition system, with the objects as states and the morphisms out of a state as the moves available there, it is a state that no sequence of challenges moves away from, which is the reading of $\dagger$ as storage that survives repeated use; the enrichment records how well the witness withstands the sequence. For $!A$ the same holds with the word replaced by a multiset, so the record of what has been used is kept but its order is not.

6 Order and sequentiality

Section 5 left the two modalities side by side and unrelated. What should relate them is the statement that whatever may be reused in parallel may be reused in sequence: $!A$, which is a comonoid for $\otimes$, ought to be a comonoid for $\triangleleft$ as well. Nothing in the definitions of the two tensors gives that, and a comonoid structure does not transport from one tensor to another without a map between them. The map is ser , which sends (u, v) to (u, const_v) and so makes the continuation ignore the challenge it is handed.

The substitution connective has been arrived at here by symmetry from Blass's $\rtimes$. Blass called his a semicolon, to suggest sequential composition, which is the idea the substitution connective is after too. There is also a linear logic calculus built to model the sequential update of state, the *Linear Logic Model of State* (LLMS) of [6, 7], and its contexts carry exactly this kind of order: a context is built by a comma and by a semi-colon, the semi-colon putting everything on its left below everything on its right, and one structural rule allows a context to be given more order than it has. Its own models put the order elsewhere. The connective of [6] has the shape of the Dialectica tensor with the predicate taken pointwise, and so is commutative; the version that models LLMS itself keeps that shape and moves the order into the values, taking them over a three-element poset and combining them with the lazy conjunction of computer science. So the order may sit in the values or in the shape, and $\triangleleft$ takes the second route.

The map ser comes from that calculus, and so does the map that should be asked for in its place. Reading the comma as $\otimes$ and the semi-colon as $\triangleright$, the modalities are governed by rules of the shapes

$$\frac{\Gamma \vdash A \quad \Delta \vdash B}{\Gamma, \Delta \vdash A \otimes B} \otimes R \qquad \frac{\Gamma[A, B] \vdash C}{\Gamma[A \otimes B] \vdash C} \otimes L$$

$$\begin{array}{c}
\frac{\Gamma \vdash A \quad \Delta \vdash B}{\Gamma; \Delta \vdash A \triangleright B} \triangleright R \quad \frac{\Gamma[A; B] \vdash C}{\Gamma[A \triangleright B] \vdash C} \triangleright L \\
\frac{\Gamma[!A, !A] \vdash C}{\Gamma[!A] \vdash C} !\text{Contr} \quad \frac{\Gamma[\dagger A; \dagger A] \vdash C}{\Gamma[\dagger A] \vdash C} \dagger\text{Thread} \quad \frac{\Gamma[\dagger A] \vdash C}{\Gamma[!A] \vdash C} !\text{Ser}
\end{array}$$

so that $!$ contracts along the comma and $\dagger$ along the semi-colon, which is Proposition 12(1) and (2) read as rules, and $!\text{Ser}$ says that what may be contracted along the comma may be contracted along the semi-colon, which is Proposition 12(3). All three are primitive rules there, and $!\text{Ser}$ is the one that Corollary 15 below interprets. Beside them stands the only rule of the calculus that looks at the order of a context, which is structural:

$$\frac{\Gamma' \vdash A}{\Gamma \vdash A} \text{Ser} \quad \text{if } \leq_{\Gamma} \subseteq \leq_{\Gamma'} \text{ on the same formula occurrences,}$$

a context being readable as a partially ordered multiset of its formula occurrences, with the comma leaving the two parts incomparable and the semi-colon putting everything on the left below everything on the right.

The instances of Ser give exactly four entailments, and two of them are the maps wanted here:

$$\frac{\Delta[A; B] \vdash C}{\Delta[A, B] \vdash C} \text{ gives } \text{ser}: A \otimes B \multimap A \triangleright B,$$

$$\frac{\Delta[(A, C); (B, D)] \vdash E}{\Delta[(A; B), (C; D)] \vdash E} \text{ gives } \text{int}: (A \triangleright B) \otimes (C \triangleright D) \multimap (A \otimes C) \triangleright (B \otimes D),$$

the other two being the weak distributivities obtained from $\Delta[(A, B); C] \vdash D / \Delta[A, (B; C)] \vdash D$ and its mirror image. So ser is where the calculus lets order be added to an unordered context, and int is where it lets two ordered contexts running side by side be resequenced.

It is int and not ser that a model should be asked for. The orders that contexts carry are the series-parallel ones, the least class containing the point and closed under disjoint union and ordinal sum, and Bechet, de Groote and Retoré [2] show that the inclusion of one such order in another is completely axiomatised by the four inferences above and, once the two constructors share a unit and the comma is commutative and associative, by the middle interchange alone. So ser and the two weak distributivities are instances of int , the first with B and C taken to be I , and one natural transformation is the whole of what the order costs.

Proposition 13. *There are morphisms in $\mathbf{Dial}_L(\mathbf{Set})$, natural in all arguments,*

$$\text{ser}_{A,B}: A \otimes B \longrightarrow A \triangleleft B, \quad \text{int}_{A,B,C,D}: (A \triangleleft B) \otimes (C \triangleleft D) \longrightarrow (A \otimes C) \triangleleft (B \otimes D).$$

Proof. For **ser**, send the witness (u, v) to (u, const_v) and a challenge (x, y) to itself; the two predicates are then literally the same element $\alpha(u, x) \otimes \beta(v, y)$ of L . For **int**, send the witness $((u, \varphi), (w, \psi))$ to $((u, w), (x, z) \mapsto (\varphi x, \psi z))$ and a challenge $((x, z), (y, t))$ to $((x, y), (z, t))$; both predicates evaluate to $\alpha(u, x) \otimes \beta(\varphi x, y) \otimes \gamma(w, z) \otimes \delta(\psi z, t)$, up to the commutativity of $\otimes$ in L . Naturality is naturality of the corresponding maps of shapes. $\square$

Theorem 14. $(\mathbf{Dial}_L(\mathbf{Set}), I, \otimes, \triangleleft)$ is a normal duoidal category: a symmetric monoidal closed structure $(\otimes, -\circ, I)$ and a second monoidal structure $(\triangleleft, I)$ on the same unit, related by the monoidal natural transformation **int**.

Proof. Theorem 3 and Propositions 8 and 13 supply the data, and the coherence conditions are inherited from **Poly**, where $(y, \otimes, \triangleleft)$ satisfies the same axioms, since $|-|$ is faithful and carries all the structure in question. The map **ser** is the instance of **int** with B and C taken to be I . $\square$

Corollary 15. Every comonoid for $\otimes$ in $\mathbf{Dial}_L(\mathbf{Set})$ is a comonoid for $\triangleleft$, with the same counit and with comultiplication $C \rightarrow C \otimes C \xrightarrow{\text{ser}} C \triangleleft C$. In particular $!A$ is a comonoid for $\triangleleft$, its comultiplication sending u to (u, const_u) and adding multisets backwards.

Proof. The counit is available because the two tensors share the unit, and coassociativity of the composite follows from naturality of **ser** together with the coherence of the two monoidal structures. $\square$

This discharges the debt of Section 5, and it is what an LLMS-category has to require of a model by hand: there the transfer is imposed, by asking for a monic **ser** that makes $!A$ a comonoid for “before”, and here it comes with the axioms.

Remark 16. The closing section of [8] asks for two things: a generalisation beyond **Poly** of the duoidality of the composition product over the parallel one, which Shapiro and Spivak use to model compositional dependence [13], and an account of how the known structures on $\mathbf{Dial}(\mathbf{Set})$ carry over to homogeneous polynomials over other bases. Theorem 14 is the first of these for homogeneous polynomials over a lineale.

Remark 17. **Poly** carries a normal duoidal structure of its own, $(\otimes, \triangleleft)$, and a theory has been built on it as a language for compositional dependence [13]. What the logical side adds to that is a reason for those axioms and no others, by the completeness result used above. A normal duoidal category is not one choice of axioms for “this and then that” among several: nothing further can be asked of the two tensors that the orders of a context do not already generate.

Remark 18 (what the second name buys). Nothing so far needs the word duoidal: the content is that $\triangleleft$ carries the predicates and that **ser** and **int** are Dialectica morphisms. Three things come with the name. The first is coherence, used in the proof above. The second is a free model: the free category on one generator with these axioms has for its objects the finite posets built from the point by disjoint union and by join, which are the series-parallel orders. The third is Corollary 15, which needs only the shared unit and the naturality of **ser**. What the name does not buy is the closure of $\otimes$, the lineale or the predicate, none of which is part of the axioms.

7 Conclusions

A Dialectica object has a shape and a predicate, and the comparison with polynomial functors separates them. The shape records the witnesses, the challenges and the ways one may depend on the other; the predicate records which witness withstands which challenge. The functor $|-|: \mathbf{Dial}(\mathbf{Set}) \rightarrow \mathbf{Poly}_{\text{mono}}$ keeps the shape and forgets the predicate, and it is faithful without being full, which is the precise sense in which the polynomial is what remains visible of a proposition.

All the interesting connectives that **Poly** carries on monomials were already Dialectica connectives more than 30 years ago. They come from one piece of work, the thesis [10], and from the two papers that preceded it [4, 9]: two constructions, two categories, each monoidal closed in its own right, each connective given with its predicate as part of a model of linear logic. Table 1 is the summary of the comparison. Read across a row and one sees a connective, its witnesses and its challenges, and then the name the polynomial literature has for the shape that is left when the predicate is dropped.

Substitution is the row that looks like an exception, but it is not one. The operation is old, being composition of polynomials; what was missing on the Dialectica side was a logical reading for it, and it has one here. A witness of $A \triangleleft B$ answers a challenge to A with the witness of B to be confronted next, and the two predicates are combined in the order the play imposes. The connective was implied by the constructions before it was written down, since the passage from no dependence $X \times Y$ to dependence on both sides $X^V \times Y^U$ already ran through them in the challenge coordinate, and negation puts the same passage in the witness coordinate. Once $\triangleleft$ is there, $\dagger$ is its comonoid and the passage from reuse in parallel to reuse in sequence is a consequence rather than an assumption, showcasing old work of Bechet-de

Groote-Retoré as well.

What none of the polynomial vocabulary can say is which witness withstands which challenge, and that is what makes a proposition out of a pattern of interaction. The lesson is not that Dialectica constructions can be replaced by polynomial ones. It is that the polynomial exposes one component of a structure Dialectica has had from the beginning, and exposing it makes the patterns of dependence easier to recognise and to organise. What it cannot do is tell two propositions apart when they share an interaction pattern and differ in their predicates.

Two companion notes take up what is left out here. One asks what a lens obliged to respect a predicate can do that a plain lens cannot, and is written for the bidirectional-transformations community. The other asks what has to be required of a monad on the challenge coordinate, and which functional interpretations the answer then covers.

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